

YappChat — Development Plan

Last Updated: 2026-06-30 · Phase: Implementation · Snapshot date: 2026-06-28 dashboard

This file is the **editable source of truth**. Change any step below, then run `pnpm devplan:pdf` (or `node scripts/build-devplan-pdf.mjs`) to regenerate `specs/DEVELOPMENT-PLAN.pdf`. Flip a step's status by changing its marker: `[ ]` → not started · `[~]` → in progress · `[x]` → done. The live, always-current status lives in the memory dashboard — this is a planning snapshot.

▶ START HERE

- **Where I am:** Phase 1 — shipping the social slice (auth + communities + chat). Foundation (Phase 0) is mostly built.
- **What's deployed:** Only the WS engine (ws.wxpers.com). `apps/web` is **not deployed yet** and is **untracked in git**.
- **Single next action:** UI-smoke the 2026-06-27 work (AI-enabled space + sidebar badges + community CRUD), then **commit** `apps/web` and deploy the slice.
- **Nearest blocker:** Spec **018 (Contacts & DMs) is unscoped** but sits on the Phase 1 path — run `buildscope / createspecs` for it before Phase 1 can fully close.

Status at a glance

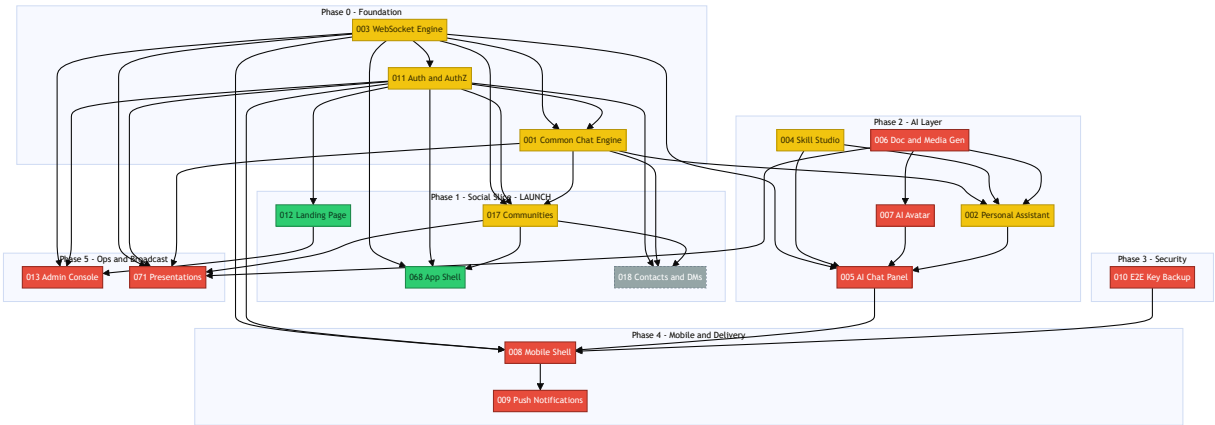
Spec	Title	State	Remaining (high-level)
003	WebSocket Engine	 ~88%	Capacity monitoring + RedisBroker (horizontal scale).
011	Auth & Authorization	 ~50%	Magic-link, device registry, AuthGate/account UI.
001	Common Chat Engine	 ~22%	23 external plugins, video, E2E, history/search.
002	Personal Assistant	 ~40%	Monitoring loop, OAuth, subagents. Migrations 0003–0004 pending.
004	Agent & Skill Studio	 ~50%	Sandbox runner, metrics, Archie assistant. Migration 0002 pending.
017	Communities	 ~28%	Discovery, native messaging, translation, RAG.
012	Public Landing Page	 100%	Done & verified.

Spec	Title	State	Remaining (high-level)
068	Authenticated App Shell	● 100%	Done & verified live.
005	AI Chat (Panel)	● 0%	Not started.
006	Document & Media Gen	● 0%	Not started.
007	AI Avatar	● 0%	Not started.
008	Mobile Shell & Packaging	● 0%	Not started.
009	Push Notifications	● 0%	Not started.
010	E2E Key Backup & Recovery	● 0%	Not started.
013	Admin Console	● 0%	Not started (scoped).
018	Contacts & Direct Messages	○ 0%	Not yet scoped.
071	Presentation (LiveKit)	● 0%	Not started (scoped).

Legend: ● built · ● partial · ● not started · ○ not scoped.

Dependency map

How the specs depend on each other, grouped by build phase. Build foundation-up: an arrow **A --> B** means **B needs A first**.



Critical path — the specs that unblock the most:

- **003 → everything realtime, 011 → every authed surface, 001 → every message.** These three are the bedrock; keep them healthy first.
- **002 (PA) is the glue** — it consumes 004/006/007 and outputs into 005. Sequence that cluster together, not piecemeal.
- **008 (Mobile) has a 4-way gate** (001 + 003 + 010 + 011) — it cannot start until 010 exists.

Phase 0 — Foundation (Identity · Transport · Message Bus)

Bedrock everything imports. Mostly built — finish only what blocks later phases.

- [~] **003 WebSocket Engine** — server/auth/broker/heartbeat/replay/presence/client-lib done. Remaining: capacity monitoring (70/90% alerts), RedisBroker (defer until >1 WS task).
- [~] **011 Auth** — core model, email/pw, refresh rotation, system flags, SSO/OIDC, org members done. Remaining: magic-link, device registry, AuthGate/account UIs.
- [~] **001 Chat Engine (core)** — tables + membership core + `/messaging` UI done & verified. Defer external plugins/video/E2E/search to their own phases.

Exit gate: auth + WS + `conversationmembers` core live-verified end-to-end. (*Largely met.*)

Phase 1 — Social Slice (current launch focus)

The slice going online first: people can sign in, join communities, and chat.

- ☒ **012 Landing Page** — done & verified.
- ☒ **068 App Shell & Dashboard** — done & verified.
- [~] **017 Communities** — spaces + join/invite/moderation + FR-019 support-AI done. Remaining for launch-quality: discovery/directory, native messaging, presence/typing.
- ☐ **018 Contacts & DMs** — ⚠ **scope this first** (`buildscope` / `createspecs`), then build the contacts graph + DM gating.
- ☐ **Launch ops:** UI-smoke today's work → commit `apps/web` → Design→Implementation lifecycle transition → deploy (app domain, prod secrets: `DATABASE_URL`, SES, Google SSO, `SITE_URL`, matching `WS_INTERNAL_SECRET`).

Exit gate: slice smoke-tested, committed, and deployed online.

Phase 2 — AI Layer

The differentiator: the Personal Assistant, its studio, its chat surface, its face. Tightly interdependent — build as one cluster.

- [~] **004 Skill Studio** — skill CRUD, schema editor, code-gen, agent templates done. Remaining: sandbox runner, metrics, Archie.
- [~] **002 Personal Assistant** — provider registry, sessions/SSE, skill-invocation runtime done. Remaining: PA channel/avatar, monitoring loop, calendar/email OAuth, subagents. (*Apply pending migrations 0003-0004.*)
- ☐ **006 Document & Media Generation** — job queue, PDF/Excel/PPTX, image gen, templates.
- ☐ **007 AI Avatar** — avatar library, `avatarconfigs`, animated state machine driven by `pa.status`.
- ☐ **005 AI Chat Panel** — slide-in panel: streaming SSE, rich content, voice, attachments, studio handoff.

Exit gate: PA answers in-chat (005) with a live avatar (007) and can generate a document (006).

Phase 3 — Security & Compliance

- ☐ **010 E2E Key Backup & Recovery** — Argon2id + XChaCha20-Poly1305 backup, 24-word recovery codes, QR cross-device pairing, group session-key recovery. Server never sees plaintext.

Exit gate: key-recovery flow round-trips on a fresh device without the server seeing plaintext.

Phase 4 — Mobile & Delivery

Gated: 008 needs 001 + 003 + 010 + 011; 009 needs 008.

- ☐ **008 Mobile Shell & Packaging** — Expo + EAS, mount the React app, `SecureKeyStore`, native video, deep linking.
- ☐ **009 Push Notifications** — APNs/FCM/Web Push fanout, silent-push-then-fetch for E2E, quiet hours, deep-link tap targets.

Exit gate: app installs on a device and receives a push.

Phase 5 — Operations & Broadcast

- ☐ **013 Admin Console** — staff control plane at `/admin`: access gate, feedback inbox, health dashboard, audit-log viewer, role management, cross-spec config.
- ☐ **071 Presentations (Webinars)** — LiveKit screen-share + narration, GROQ live captions/translation, hand-raise, access-scoped replay, 100-attendee cap.

Exit gate: staff console live; a webinar can be hosted end-to-end.

How to regenerate the PDF

```
pnpm devplan:pdf          # or: node scripts/build-devplan-pdf.mjs
```

Writes `specs/DEVELOPMENT-PLAN.pdf` with the diagram rendered as a real graphic. Edit any step above first — the PDF always reflects the current markdown. The script is offline (mermaid is injected from `node_modules`, no network needed).